

When does a three-level readout help?

Leakage flags, erasure qubits and the qg T1-aware decoder

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Draft of 29 September 2026; revised 30 September 2026

Abstract

Transmon qubits leak to their second excited state, and a three-level readout can flag it. We ask whether this helps the qg T1-aware decoder, a minimum-weight matching decoder whose weights follow the final data readout and which a register-mean polar-bias witness switches on only when relaxation dominates [1, 2]. Six simulated tests on a circuit-level surface-code memory, each with its criterion fixed before the reported run, give a consistent answer. (i) When a leaked qubit keeps its value, erasing flagged qubits makes every decoder worse, and the whole qg gain ($1.68\times$) comes from the qubit-level reweighting. (ii) When leakage scrambles the value, the flag and the reweighting combine better than either alone: $1.68\times$ over the best leakage-aware decoder at distance 3. (iii) A Bayesian weight for the flag, a flip probability $a/(1+a)$ with a the leak asymmetry between $|0\rangle$ and $|1\rangle$, contains both cases, is never worse than ignoring leakage, and is not uniformly better than erasure (one of four predictions fails). The combined gain grows with distance, from $1.48\times$ to $1.77\times$. (iv) A coherent three-level simulation of a diabatic CZ gives $a = 0$ and a leaked qubit that keeps its value and still flips its ancilla with probability 0.97. For such transmons the flag is useless, erasure hurts, and the qg gain ($1.81\times$ at $d = 3$, $2.02\times$ at $d = 5$) needs no qutrit readout. (v) On erasure qubits, which herald T1 decays as they happen, a naive combination with qg is harmful when heralding is good ($0.46\times$ at heralding efficiency $h = 0.99$); scaling the prior of unheralded decays by $1 - h$, a rule then tested with its own pre-registration, removes the loss and keeps a gain of 1.41 – $1.68\times$ at $h = 0.5$. All results are simulations, reproduced by scripts and pinned by regression tests.

1 The question

The qg unit describes a qubit by its polar bias $qg_Z = \langle \sigma_z \rangle$ and related quantities [1]. In a surface-code memory, relaxation (T1) biases the final data readout towards 0; a companion preprint [2] uses this to reweight the matching graph shot by shot (a data qubit read as 1 has not decayed, so its decay faults are discounted), and switches the reweighting on only when the register-mean qg_Z of a calibration batch, divided by the detector rate, exceeds 1. We call this the qg decoder.

Superconducting qubits also leak out of the computational space, mainly to $|2\rangle$, and leakage is a known threat to error correction [3–5]. A three-level readout can see it, and leakage-aware decoders treat flagged qubits as erasures [4, 6, 7]. Before extending qang beyond qubits we asked a narrow question: does the qg decoder gain anything from a three-level readout, beyond what standard leakage handling already gives? The rule agreed in advance was to extend qang only if it does.

2 Setup

Memory and noise. A rotated surface code in a Z-memory experiment, distance $d = 3$ with 3 rounds or $d = 5$ with 5 rounds, simulated shot by shot at circuit level. Noise: two-qubit

Table 1: Logical error, first model ($a = 0$, value kept), $d = 3$, 60 000 shots per regime.

regime (witness)	standard	erasure	qg + erasure	qg, no flags
leakage + T1-dominated (1.68)	0.0156	0.0165	0.0100	0.0093
leakage + mixed (1.27)	0.0099	0.0106	0.0089	0.0084
leakage + depolarizing (0.34)	0.0032	0.0036	0.0036	0.0032
strong leakage, depolarizing (0.46)	0.0021	0.0032	0.0032	0.0021

depolarizing p_2 after each CNOT, amplitude damping γ per layer (twice during measurement), readout flips q . Two regimes: T1-dominated ($p_2 = 0.0005$, $\gamma = 0.003$, $q = 0.002$) and depolarizing ($p_2 = 0.004$, $\gamma = 0.0005$, $q = 0.002$).

Leakage. After a CNOT a data qubit leaks with probability $\ell = 0.002$ if it is in $|1\rangle$ and $a\ell$ if it is in $|0\rangle$. A leaked control kicks its ancilla with probability κ (1/2 unless stated), does not decay, and returns with probability r per layer. Three choices define a model: the asymmetry a , the value it returns to (always 1, or random), and how a two-level readout assigns $|2\rangle$ (as 1, or at random). The final three-level readout flags every qubit still leaked.

Decoders. All use minimum-weight perfect matching (PyMatching [8]) on the same detector graph: *standard* (leakage ignored); *erasure* (the final-readout edges of a flagged qubit get probability 1/2, or, where stated, all its edges); *Bayesian weight* (Section 3.3); *qg* (the T1 reweighting); and the combinations qg + erasure and qg + Bayesian weight. Decoders are compared on the same shots with a paired (McNemar) z .

Protocol. Each test states its criterion in the script before the run. For the last five, script and criterion were committed to the public repository before the reported run (commits 3fa4d63, daf0b6f, 092580f, e877a72, 68d8d00), and the code was debugged on a different seed.

3 Results

3.1 A leaked qubit that keeps its value: a negative result

In the first model ($a = 0$, return to 1, $|2\rangle$ read as 1, $r = 0.05$) we compared qg + erasure with erasure. The criterion ($1.3\times$, $z \geq 3$, never worse) formally passed with $1.65\times$ ($z = +14.8$), but the pass was an artefact (Table 1). The erasure baseline, which erased a flagged qubit’s whole history, was worse than ignoring leakage in every regime ($z = -3.8$ to -6.0). The qg decoder *without* any flag beat standard by $1.68\times$ ($z = +14.7$), and adding flags to it made it worse (z between -2.8 and -4.9). Erasing only the last layer, a variant added after the first run, was less harmful but still worse than standard. The reason is simple: a qubit that leaks only from $|1\rangle$ and returns to $|1\rangle$ is read correctly as 1, so declaring it erased discards a right value. We recorded a negative result.

3.2 Leakage that scrambles the value

An exploratory rerun suggested that the flag helps when the leaked qubit loses its value. We then fixed a second model and a stricter criterion before the run: leakage from $|1\rangle$ (ℓ) and from $|0\rangle$ ($\ell/2$), return to a random bit, $|2\rangle$ read at random, and per-round leakage flags with efficiency 0.8. The criterion required the qg decoder with flags to beat the *better* of two leakage-aware decoders by $\geq 1.3\times$ with $z \geq 3$ in some regime, never lose with $z \leq -3$, and the flags to improve qg in every regime.

Table 2: Logical error in the T1-dominated regime, 60 000 shots per point. p_{flag} from Eq. (1).

d	model	p_{flag}	standard	erasure	Bayesian	qg	qg + erasure	qg + Bayesian
3	M_0	0.002	0.0154	0.0163	0.0153	0.0091	0.0097	0.0093
3	$M(0)$	0.002	0.0177	0.0171	0.0161	0.0104	0.0105	0.0099
3	$M(0.25)$	0.200	0.0182	0.0175	0.0170	0.0119	0.0112	0.0110
3	$M(0.5)$	0.333	0.0189	0.0181	0.0180	0.0137	0.0122	0.0121
3	$M(1)$	0.500	0.0202	0.0192	0.0192	0.0171	0.0140	0.0140
5	M_0	0.002	0.0100	0.0103	0.0100	0.0045	0.0049	0.0047
5	$M(0)$	0.002	0.0118	0.0112	0.0109	0.0055	0.0053	0.0052
5	$M(0.5)$	0.333	0.0141	0.0131	0.0131	0.0085	0.0073	0.0074
5	$M(1)$	0.500	0.0170	0.0153	0.0153	0.0125	0.0098	0.0098

Both parts passed. In the T1-dominated regime qg + flags reached 0.0106 against 0.0177 for the best leakage-aware decoder ($1.68\times$, $z = +15.3$), and $1.28\times$ in the mixed regime; the flags improved qg in all four regimes ($z = +4.0$ to $+8.9$). The combination is more than the product of its parts: flags alone gave $1.07\times$ over standard, the reweighting alone $1.45\times$, together $1.78\times$. A plausible mechanism is that the reweighting trusts the final data bits, and a leaked qubit’s final bit is random, so erasing it removes a misleading input. A second seed gave $1.43\times$ ($z = +11.3$). Per-round detection added little at $d = 3$: with the final readout alone the gain was $1.39\times$.

3.3 A rule for the flag

The two results disagree because both erase the flag regardless of where the leak came from. If a qubit leaks from $|1\rangle$ with probability ℓ and from $|0\rangle$ with $a\ell$, and is seen in $|2\rangle$, its value before leaking was 1 with posterior $1/(1+a)$ (uniform prior). The Bayesian weight therefore reports the flagged qubit as 1 and gives its final-readout edges the flip probability

$$p_{\text{flag}} = \max\left(\frac{a}{1+a}, q\right), \quad (1)$$

with a neutral T1 weight, since a leaked qubit does not decay. At $a = 0$ it trusts the flag’s value; at $a = 1$ it is plain erasure. The first model is the case $a = 0$, where the rule reduces to the standard decoder.

We tested four predictions over five models (M_0 , the first model; $M(a)$ with $a = 0, 0.25, 0.5$ or 1 , random return and random readout; $M(0.5)$ is the second model), both regimes at $d = 3$ and four models at $d = 5$ (Table 2, Fig. 1). **P1** (never worse than standard) passed: z from 0.0 to $+9.0$, while erasure lost to standard in M_0 ($z = -5.7$). **P2** (beats erasure for $a \leq 0.25$ in both regimes) failed: it held in the T1-dominated regime ($z = +6.1, +7.0, +3.9$) but not in the depolarizing one ($+2.9$ for M_0 , -0.4 for $M(0.25)$), and at $a = 0.5$ depolarizing erasure was slightly better ($z = -2.2$); the differences are at most 5%. A leaked qubit also scrambles its ancillas for several rounds, which the posterior on its value ignores. **P3** (qg + Bayesian weight never worse than qg + erasure) passed. **P4** (a qg + flag gain of $\geq 1.3\times$ at both distances in $M(0.5)$) passed, and the gain grew: $1.48\times$ ($z = +12.5$) at $d = 3$, $1.77\times$ ($z = +13.6$) at $d = 5$, and 1.55 – $2.11\times$ across the four $d = 5$ models. As pre-registered, with P2 failing the rule counts as partly confirmed.

3.4 Where a transmon sits

The parameters of the models above were chosen, not derived. We derived them for a transmon from a coherent simulation: two transmons truncated at three levels (anharmonicity $-2\pi \cdot 0.3$ GHz, coupling $2\pi \cdot 10$ MHz) and a diabatic flux-pulse CZ through the $|11\rangle \leftrightarrow |20\rangle$ resonance [9], with the hold time optimised for gate fidelity after virtual-Z corrections. The gate reached an

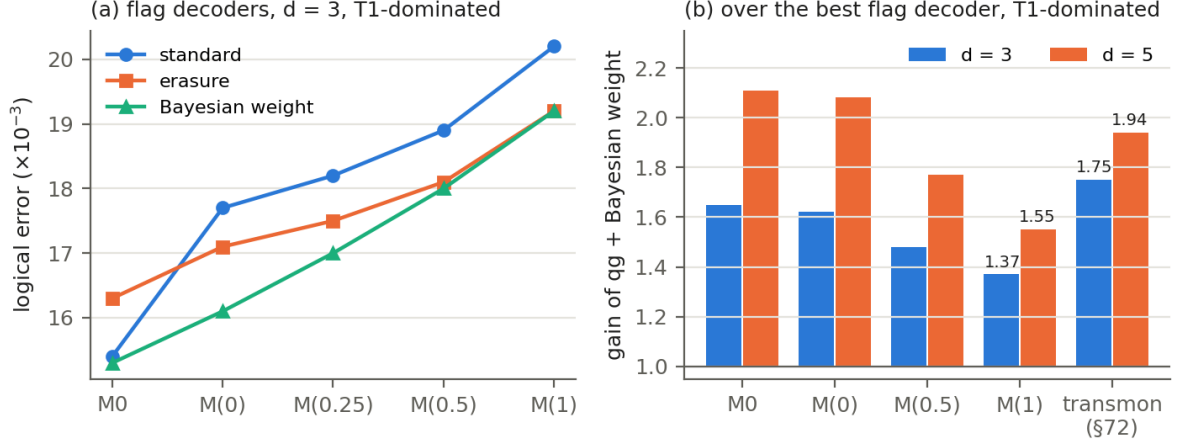


Figure 1: (a) Logical error of the three flag decoders across the leakage models, $d = 3$, T1-dominated regime. (b) Gain of qg + Bayesian weight over the better of erasure and Bayesian weight, at $d = 3$ and $d = 5$, including the transmon channel of Section 3.4.

Table 3: Logical error with the leakage channel derived for a transmon CZ, 60 000 shots per point.

d	regime (witness)	standard	erasure	Bayesian	qg
3	T1-dominated (1.68)	0.0153	0.0163	0.0153	0.0084
3	depolarizing (0.34)	0.0031	0.0035	0.0030	0.0031
5	T1-dominated (2.30)	0.0079	0.0084	0.0079	0.0039

average fidelity of 0.99967 with leakage 5.4×10^{-4} from $|11\rangle$ and exactly zero from the other inputs, so $a = 0$. This is structural: the exchange coupling conserves the number of excitations, and only $|11\rangle$ can reach $|20\rangle$. A leaked data qubit gives its ancilla a conditional phase of 0.89π , so in a CNOT it still flips the ancilla with probability $\kappa = 0.97$, almost as if it were in $|1\rangle$; and it decays back to $|1\rangle$ at twice the T1 rate. The leaked transmon is nearly invisible to its stabilizers and keeps its value: the M_0 corner of the map.

With this channel in the memory (and $|2\rangle$ read as 1), all four predictions made before the run passed (Table 3). The Bayesian weight tied with standard ($1.001\times$, $1.011\times$, $0.998\times$; $|z| \leq 1.4$), erasure hurt ($z = -7.2$, -4.9 , -5.7), and the qg decoder without flags gained $1.81\times$ ($z = +15.3$) at $d = 3$ and $2.02\times$ ($z = +12.4$) at $d = 5$. Adding the flag to qg cost a little, because the flagged qubit’s T1 weight is set to neutral although its bit is right.

3.5 Erasure qubits

Erasure qubits, such as dual-rail transmons [10] and atoms with erasure conversion [6], herald decay events as they happen. If heralding were perfect, the final-readout bias that the qg decoder uses would carry nothing extra. We modelled this on the same memory without leakage: every data-qubit decay $1 \rightarrow 0$ is heralded with probability h at its exact location, and the erasure decoder sets that edge to probability $1/2$.

The first test predicted that the gain of qg + erasure over erasure would fall with h (it did: from $1.59\times$ and $2.04\times$ at $h = 0$ to below $1.1\times$ at $h = 0.99$) and that the combination would never be worse than erasure alone. The last prediction failed, and the failure is the finding: at $h = 0.99$ qg + erasure reached only $0.46\times$ ($z = -6.7$) at $d = 3$ and $0.22\times$ at $d = 5$. The T1 reweighting assumes every decay is unheralded; with heralds, a 0 at the final readout is mostly explained already, and the reweighting counts it twice. The witness does not catch this, because

Table 4: Gain of qg + erasure over erasure, both with unheralded decay priors scaled by $1 - h$ (paired z in brackets), 60 000 shots per point.

regime	d	$h = 0.25$	0.5	0.75	0.9	0.99
T1-dominated	3	1.50 (+11.1)	1.41 (+9.0)	1.25 (+4.2)	1.09 (+1.2)	0.92 (−0.8)
T1-dominated	5	1.93 (+9.0)	1.68 (+5.4)	1.39 (+2.8)	1.45 (+2.2)	1.00 (0.0)
mixed	3	1.19 (+4.9)	1.18 (+4.1)	1.16 (+2.6)	1.07 (+1.2)	1.12 (+1.8)
mixed	5	1.47 (+5.2)	1.30 (+2.7)	1.03 (+0.3)	0.79 (−2.0)	1.00 (0.0)

the readout bias is still there at every h .

An exploratory fix, found after that run, scales the prior of *unheralded* decays by $1 - h$ in both decoders. We then tested it with a new pre-registration: new seeds, a second (mixed) regime, five values of h , and h misestimated by about 0.1. All four predictions passed (Table 4). With the rule, qg + erasure was never significantly worse than calibrated erasure (closest point $0.79\times$, $z = -2.0$, mixed regime, $d = 5$, $h = 0.9$); it gained $1.41\times$ ($d = 3$) and $1.68\times$ ($d = 5$) at $h = 0.5$ and faded to about 1 at $h = 0.99$; calibration also improved erasure alone at high h ; and an error of 0.1 in h changed little.

4 What decides it

The six tests fit one picture. Whether a three-level readout helps is decided by two properties of the leakage channel: the asymmetry a , and whether a leaked qubit keeps its value while leaked. When $a \approx 0$ and the value is kept, as for the transmon CZ simulated here, the flag carries no new information, erasure throws away a correct bit, and the only gain is the qubit-level T1 reweighting. When leakage scrambles the value, the flag and the reweighting are complementary and their gain grows with distance. The Bayesian weight of Eq. (1) is a safe default across this range, because it reduces to the standard decoder where the flag is uninformative; it is not the best choice everywhere. For qang the conclusion is conservative: the qg quantities remain qubit quantities, and a three-level readout enters, where it helps, as an input to the qg decoder, not as a new qg. Heralded erasures are the limit where the extra information is complete: there the qg reweighting must be told how much has already been heralded, through the $1 - h$ scaling, and it fades as $h \rightarrow 1$.

5 Limitations

(i) Everything is simulation; there are no hardware data. (ii) Distances 3 and 5 only, one leakage rate, leakage after CNOTs only, and no leakage-reduction units [5]. (iii) The leakage models are classical; only the transmon gate of Section 3.4 is coherent, with one pulse shape and parameter set, three-level truncation, and a leak rate in the memory set four times above the coherent value for statistics. The partner kick at a leak event and the assignment of $|2\rangle$ to 1 are assumed. (iv) The Bayesian weight assumes a is known from calibration and a uniform prior on the pre-leak value; one prediction about it failed. (v) The erasure model heralds data-qubit decays only, at their exact location, with h known to about 0.1. (vi) Erasure decoding of leakage and leakage detection are known [4, 6, 7]; the contributions here are the test of the qg decoder against them, the Bayesian weight with its pre-registered test, the growth of the combined gain with distance, and the placement of a simulated transmon on this map, and the $1 - h$ rule for combining qg with heralded erasures.

6 Code availability

`examples/qutrit_leakage_qg.py`, `examples/qutrit_leakage_rounds_qg.py`,
`examples/qutrit_bayes_weight_qg.py`, `examples/transmon_leakage_channel_qg.py`,
`examples/erasure_qubits_qg.py` and `examples/erasure_calibrated_qg.py` in <https://github.com/Viny2030/qang> (MIT license), with regression tests. Sections 70–74 of `RESEARCH_NOTES.md` give more detail.

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